

The First Habitable Zone Earth-sized Planet from TESS: A Summary of the TOI-700 System

Gökhan Selvi

August 19, 2025

Abstract

The TESS (Transiting Exoplanet Survey Satellite) has enabled the discovery of a three-planet system orbiting the M-dwarf star TOI-700. The most remarkable planet in this system, TOI-700 d, has a radius close to Earth's size and is located in its star's habitable zone. This paper provides a summary of the observations made by TESS and the methods used to validate the planets' parameters. The purpose of this paper is to highlight the significance of this system for the TESS mission and its great potential for future telescope observations.

1 Introduction

In the field of planetary science and astrophysics, the search for small, rocky planets has gained momentum in recent years. This search has made great strides, especially thanks to space missions such as TESS (Transiting Exoplanet Survey Satellite). The main goal of TESS is to discover transiting planets orbiting bright, nearby stars. The TOI-700 system stands out as one of the most important achievements of this mission because it hosts an Earth-sized planet located in the habitable zone.

2 The TOI-700 System

TOI-700 is an M-dwarf star located approximately 101.9 light-years away in the constellation Doradus, one of the constellations TESS continuously observes. TESS detected signals of three different planets transiting the star's light curve. These planets, listed from the inside out, are: TOI-700 b, TOI-700 c, and TOI-700 d. The outermost planet, TOI-700 d, attracts special attention because it is located within its star's habitable zone. This zone refers to the distance where liquid water can be present on a planet's surface. The planet's radius is approximately 1.19 times Earth's radius.

3 Method and Validation

TESS data was initially used to identify potential candidates. However, additional observations were needed to validate these candidates as planets. In this process, TESS's own Data Validation (DAVE) tool and ground-based telescope observations were used. While determining the physical properties of the star, the following basic equation showing the relationship between the star's effective temperature, luminosity, and radius was utilized:

$$L = 4\pi R^2 \sigma T_{eff}^4 \quad (1)$$

In this equation, L represents the star's luminosity, R the star's radius, σ the Stefan-Boltzmann constant, and T_{eff} the effective temperature. Spectroscopic data enabled the accurate calculation of the planets' true radii.

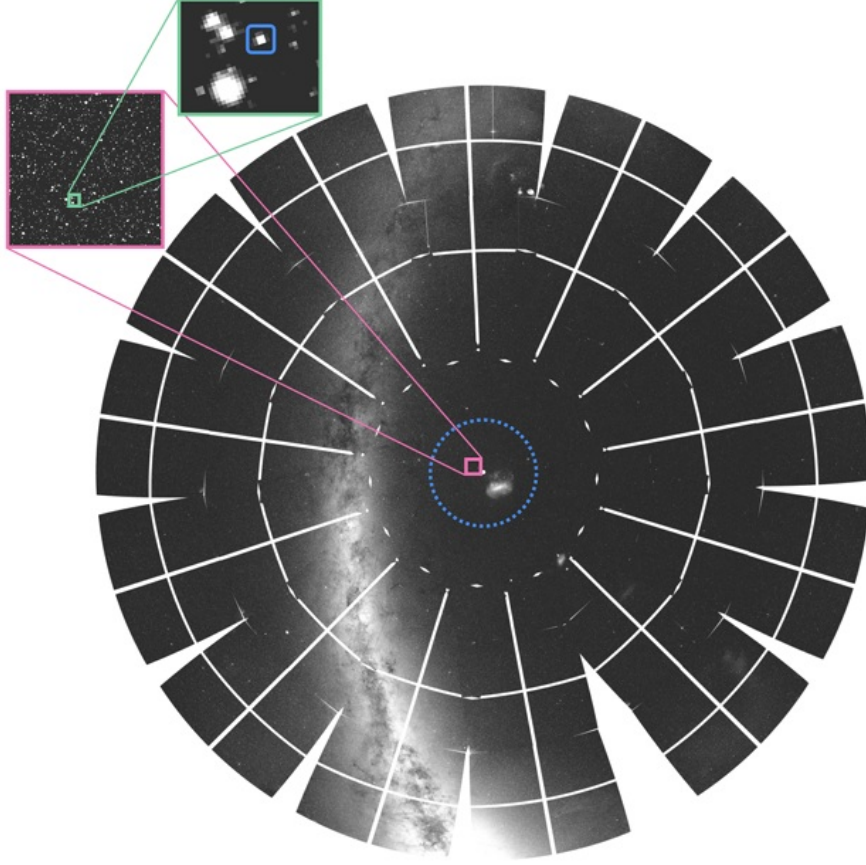


Figure 1: Position of TOI-700 within TESS’s field of view.

Parameter	TOI-700 b	TOI-700 c	TOI-700 d
Radius ($R_{\oplus}$)	1.0	2.63	1.19
Period (days)	9.98	16.05	37.43
Habitable Zone	No	No	Yes

Table 1: TOI-700 System Planet Parameters